

Spotify Popularity Algorithm

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Project Definition

- ❖ The Big Question: We all listen to Spotify, but how does a song actually become a "hit"? The Popularity Score is like a black box, and artists are desperate to know the secret!
- ❖ The Goal: Building a system to find this solution. I want to predict how popular a song will be before it gets a million streams, using hard data from the Spotify API.
- ❖ Our Achievement: Other than just predicting, I want to figure out which specific audio traits (like energy) are actually responsible for making a listener repeat a song.
- ❖ The Technical Approach: This is a full-stack project! I am creating a clean, professional data pipeline: pulling fresh data from the Spotify API, organizing it perfectly in a SQL database, and then letting Machine Learning models like Random Forest do the most important work.

Introduction

The Challenge: Decoding Popularity

- ❖ Music consumption is now driven by algorithms (Discovery Weekly, etc.), which makes the factors of a song's success more obscure than ever.
- ❖ The Spotify Popularity Score is the industry's critical metric, but it acts like a black box, combining plays and engagement in a way that is not publicly detailed.

Novelty:

- ❖ Aim to fill this knowledge gap by building a model to investigate which measurable audio traits (like danceability, energy, etc.) truly influence a song's viral success.

Introduction

Brief Results & Takeaways

- ❖ Integrated Data System: Successfully built a robust system by extracting track and artist data from the Spotify API, integrating it with a static audio features dataset, and combining everything into a multi-table SQLite database.
- ❖ Predictive Success: Using a Random Forest Regressor, I established a strong predictive baseline for the target popularity score.
- ❖ Evaluation: The model's performance was measured using key regression metrics, yielding an R-squared of 0.8173 and a Root Mean Squared Error (RMSE) of 10.36.

Methodology

Data Science / Collection Component:

- ❖ API Authentication: Used the Spotify Web API with Client Credentials Flow to gather public metadata.
- ❖ Track & Artist Data: Collected track-level information (name, release date, popularity) and primary artist characteristics (name, ID) for songs.
- ❖ Pivot/Adjustment: There was a 403 error preventing access to live audio features, so we successfully pivoted to using a public static dataset containing features like energy, danceability, and tempo

Methodology

Database / Integration Component:

- ❖ Data Integration Pivot: Due to API restrictions, we integrated API-collected track/artist data with a separate public static dataset containing audio features (e.g., danceability, energy, tempo).
- ❖ SQL Schema: Designed and implemented a multi-table SQLite database using Tracks, Artists, and AudioFeatures tables.
- ❖ Data Finalization: Cleaned and standardized data , then used a JOIN query to merge all sources into a single, analysis-ready

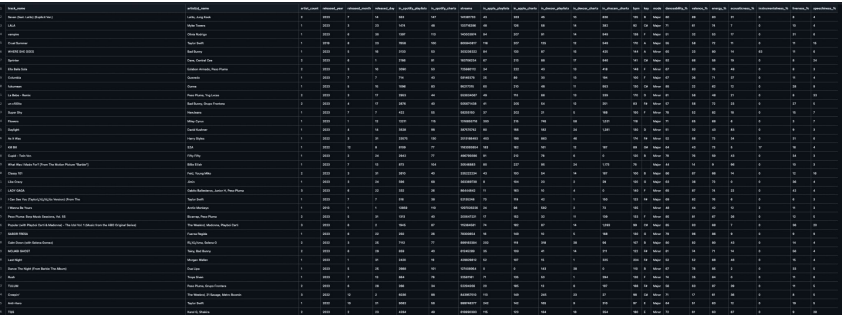
Methodology

Machine Learning Component:

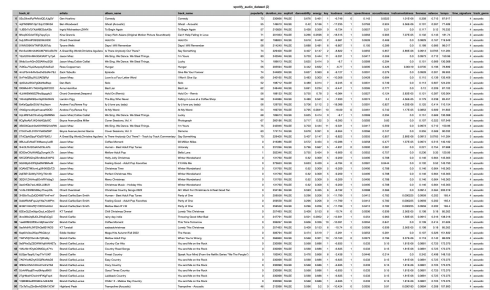
- ❖ Preprocessing & Features: Loaded the final merged data and performed feature engineering by extracting release_year and release_month from the release_date.
- ❖ Model Selection: Employed a Random Forest Regressor to effectively capture complex, non-linear relationships between audio traits and the target popularity score.
- ❖ Evaluation: Model performance was assessed using standard regression metrics: Root Mean Squared Error (RMSE) and R-squared.

Snippets of CSV Files)

- ❖ API Setup & Data Collection
- ❖ Brushed up/completed the final report
- ❖ SQL & Data Integration
- ❖ Submitted our code GitHub
- ❖ ML Learning
- ❖ Created the Slide Presentation



This image shows a snippet of a CSV file with a dark background. The columns include track_name, album_name, artist_name, and various numerical features. The data is organized in a grid-like structure with many rows and columns.



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Column	Type	Origin	Purpose in ML
popularity	Int	Original	TARGET VARIABLE (y vector)
danceability	Float64	Original	Numerical Feature
energy	Float64	Original	Numerical Feature
loudness	Float64	Original	Numerical Feature
speechiness	Float64	Original	Numerical Feature
acousticness	Float64	Original	Numerical Feature
instrumentalness	Float64	Original	Numerical Feature
liveness	Float64	Original	Numerical Feature
valence	Float64	Original	Numerical Feature
tempo	Float64	Original	Numerical Feature
duration	Int	Original	Numerical Feature
artist_popularity	Float64	Original	Numerical Feature
release_year	Int	Engineered	Feature derived from release_date
release_month	Int	Engineered	Feature derived from release_date
key	Int	Original	Categorical Feature (Treated as numerical)
mode	Int	Original	Categorical Feature (Treated as numerical)
time_signature	Int	Original	Categorical Feature (Treated as numerical)

Results

- ❖ I chose the Random Forest Regressor since it is proven in music analytics to handle non-linear data and provide robust results.
- ❖ The model was highly effective in predicting the continuous track_popularity score (range 0–100).

Metric	Value	Interpretation
R-Squared	0.8173	High Explanatory Power: Our features (audio, artist, and date) explain over 81% of the variability in a song's popularity score.
Root Mean Squared Error (RMSE)	10.36	Average Error: The model's predictions were, on average, only 10.36 points away from the actual Spotify Popularity Score.

References

❖ Articles Used:

- <https://www.scitepress.org/Papers/2024/133300/133300.pdf>
- <https://ieeexplore.ieee.org/document/9776765>